First-order approximation algorithms

Weekend 1, Friday morning

Carlos Cotrini

Friday 4 September 2026

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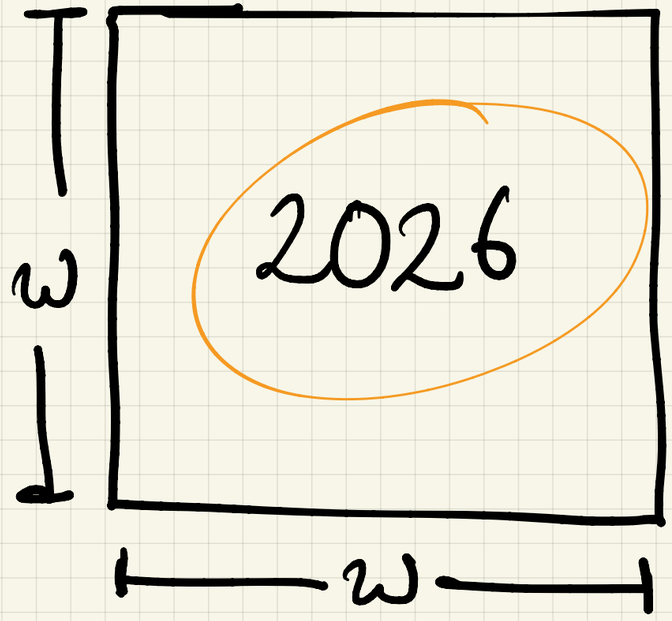
1

Section 1

Two problems, done by hand

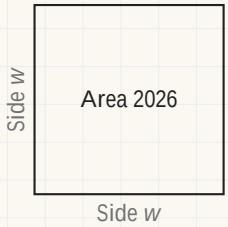


w	estimate w^2	diff: 2026 − w^2 / target − estimate	Action
45	2025	+1	↑
45.5	2070.25	−44.75	↓
45.05	2029.5	−3.5	↓

$w * w = 2026$

Heron

Calculating square roots

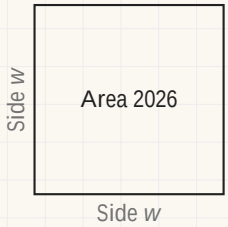


How long is the side?

Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update

Calculating square roots



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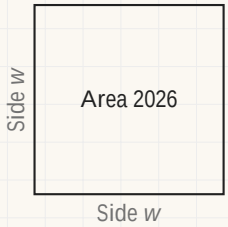
Find w with $w^2 = 2026$.

w	w^2	$l = 2026 - w^2$	Update
40	1600	$+426$	↑

Calculating square roots



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50	2500	-474	$\downarrow$

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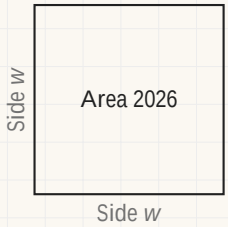

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45	2025	$+1$	$\uparrow$

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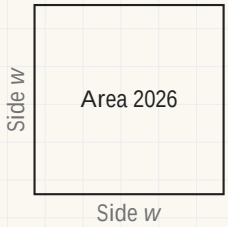

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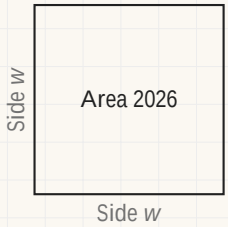

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⋮	⋮	⋮	

Nobody here computed a square root. Each row is a guess that was scored, and the score said which way to move.

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

estimate

target

w	$f(w)$	$l = 4 - f(w)$	Update
5	63	4-63=-59	↓
3			

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$I = 4 - f(w)$	Update
0	-2	$+6$	$\uparrow$

Solving an equation

Find w with $w^3 - 3w^2 + 3w - 2 = 4$.

w	$f(w)$	$I = 4 - f(w)$	Update
0	-2	$+6$	$\uparrow$
6	124	-120	$\downarrow$

Solving an equation

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0	-2	$+6$	$\uparrow$
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2	0	$+4$	$\uparrow$

Solving an equation

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0	-2	$+6$	$\uparrow$
6	124	-120	$\downarrow$
2	0	$+4$	$\uparrow$
$\vdots$	$\vdots$	$\vdots$	

The same four columns, and only f changed.

Written another way,

$$w^3 - 3w^2 + 3w - 2 = (w-1)^3 - 1,$$

so the equation asks for $(w-1)^3 = 5$: a cube root wearing a cubic.

```
def solve( f , target)
    w ← "guess"
    while |l| > 0.0001:
        l ← target − f(w)
        if l>0 :
            w ← w + 0.001 l
        else:
            w ← w − 0.001 l
    return w
```

Goal

Find w

$f(w) = target$

```
def solve( f , target, α)
    w ← "guess"
    while |l| > 0.0001:
        l ← target − f(w)
        w ← w + α l
    return w
```

Goal

Find w

$f(w) = target$

Robbins–Monro algorithm

The algorithm

```
def solve(f, target, w, alpha):
    l = target - f(w)           # the verdict: how far off, and which way
    while abs(l) > 0.001:       # not close enough yet
        w = w + alpha * l       # the update: move that way
        l = target - f(w)       # score the new guess
    return w
```

The algorithm

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def solve(f, target, w, alpha):
    l = target - f(w)          # the verdict: how far off, and which way
    while abs(l) > 0.001:       # not close enough yet
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```

The two problems of the last ten minutes are the same call, with a different `f`:

```
def square(w):
    return w**2
def cubic(w):
    return w**3 - 3*w**2 + 3*w - 2
solve(square, 2026, 40, 0.005)
solve(cubic,     4,  0, 0.05)
```

The algorithm

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def solve(f, target, w, alpha):
    l = target - f(w)          # the verdict: how far off, and which way
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    return w
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```

Everything on this slide is settled except `alpha`. That is the next section.

2

Section 2

Iterative problem solving

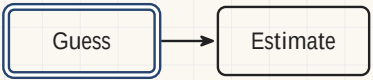
Iterative problem solving

Iterative problem solving

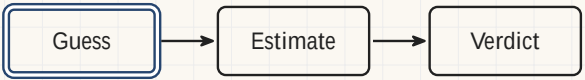
Guess A value of w, however crude.

Iterative problem solving

Guess	A value of w, however crude.
Estimate	What the guess predicts.

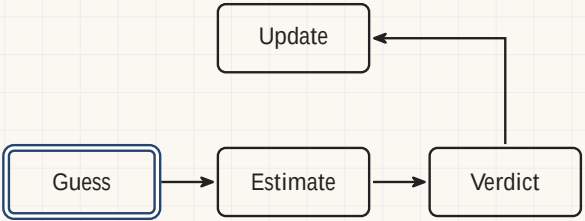


Iterative problem solving



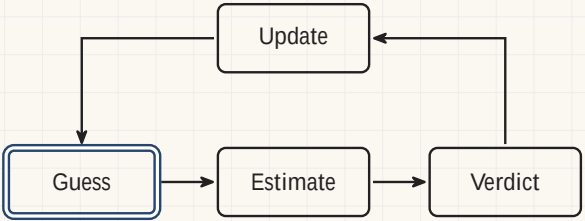
Guess	A value of w, however crude.
Estimate	What the guess predicts.
Verdict	The signed miss I.

Iterative problem solving



Guess	A value of w, however crude.
Estimate	What the guess predicts.
Verdict	The signed miss l.
Update	Move w the way the verdict says.

Iterative problem solving



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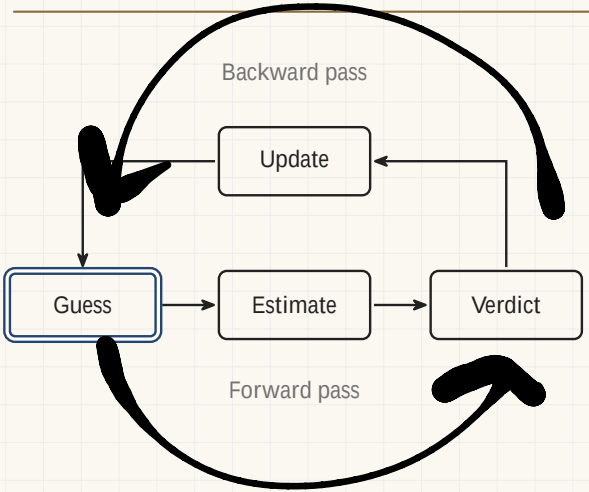

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Update	Move w the way the verdict says.

The update rule

$$w := w$$

w	w^2	I

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The update rule

$$w := w + \underbrace{l}_{\text{The verdict}}$$

w	w^2	l

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$$w := w + \underbrace{\alpha}_{\text{How far to move}} \underbrace{l}_{\text{The verdict}}$$

w	w^2	l
40.000	1600.000	426.000
42.130	1774.937	251.063
43.385	1882.286	143.714
44.104	1945.153	80.847
44.508	1980.973	45.027
44.733	2001.064	24.936
44.858	2012.234	13.766
44.927	2018.414	7.586
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45.011	2025.965	0.035
45.011	2025.981	0.019
45.011	2025.989	0.011
45.011	2025.994	0.006
45.011	2025.997	0.003
45.011	2025.998	0.002
45.011	2025.999	0.001

- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the □fth row of this block, w has stopped moving and the verdict is still shrinking.

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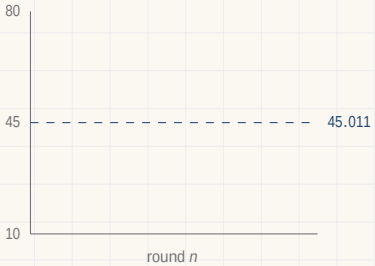
- Starting at $w = 40$, with $\alpha = 0.005$.
- The whole run, all 23 rounds of it.
- From the □fth row of this block, w has stopped moving and the verdict is still shrinking.
- $\sqrt{2026} = 45.011110$, and no square root was ever taken.

α is the step size. The letter ε is reserved: it is the stopping tolerance, and at 11:00 it becomes the width of a di□erence window.

Letting the step shrink

$w := w + \alpha_n \, l$, with $\alpha_n = 0.05/n$: the step gets smaller every round.



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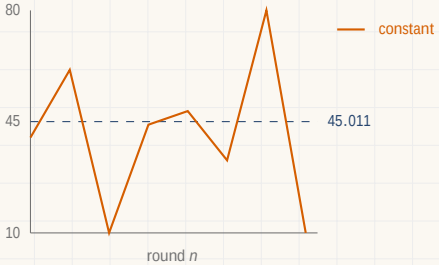

The constant run leaves the picture twice.

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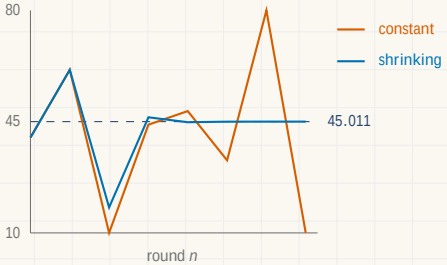

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n	α_n	w	l
1	0.0500	40.000	$+426.000$
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3	0.0167	18.008	$+1701.721$
4	0.0125	46.370	-124.155

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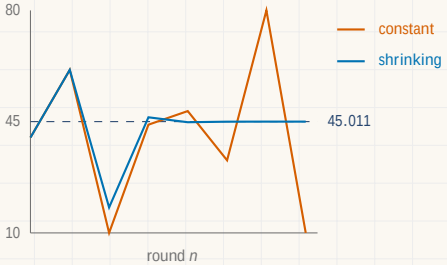

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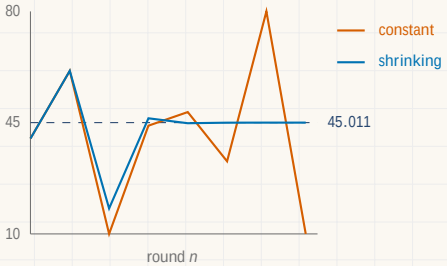

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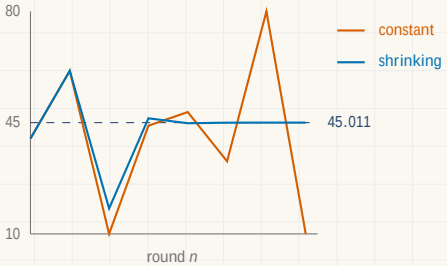

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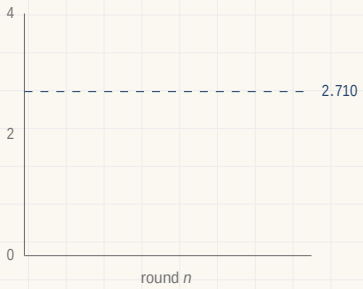
9	0.0056	45.010342	$+0.069$
12	0.0042	45.010985	$+0.011$

$\sqrt{2026} = 45.011110$. The digits arrive one at a time, and no square root was ever taken.

Letting the step shrink: the equation

The same change, on $w^3 - 3w^2 + 3w - 2 = 4$, with $\alpha_n = 0.3/n$.



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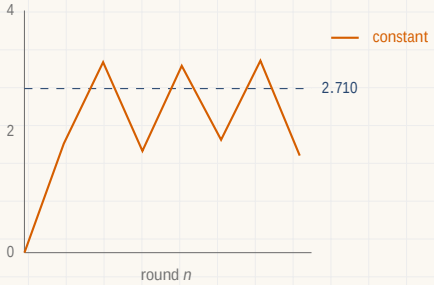

The constant run never lands.

n	α_n	w	l

Letting the step shrink: the equation

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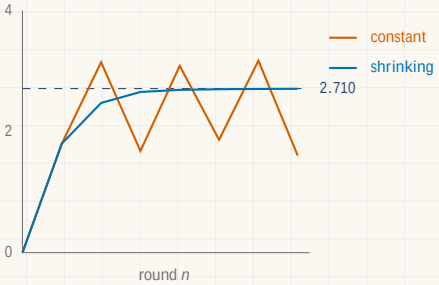

The constant run never lands.

n	α_n	w	l
1	0.3000	0.000	+6.000
2	0.1500	1.800	+4.488
3	0.1000	2.473	+1.803
4	0.0750	2.654	+0.480

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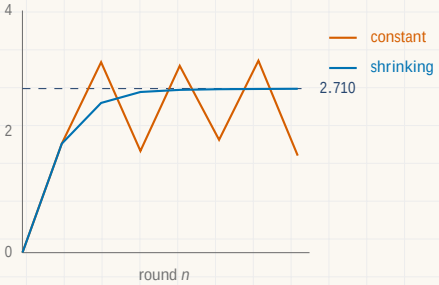

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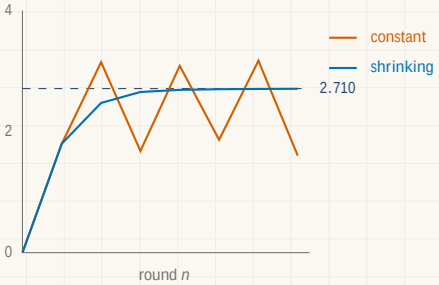
4	0.0750	2.654	+0.480
5	0.0600	2.689	+0.178
6	0.0500	2.700	+0.086

Letting the step shrink: the equation

The same change, on $w^3 - 3w^2 + 3w - 2 = 4$, with $\alpha_n = 0.3/n$.



The constant run never lands.

n	α_n	w	l
1	0.3000	0.000	+6.000
2	0.1500	1.800	+4.488
3	0.1000	2.473	+1.803
4	0.0750	2.654	+0.480
5	0.0600	2.689	+0.178
6	0.0500	2.700	+0.086
9	0.0333	2.707640	+0.020
12	0.0250	2.709048	+0.008

Letting the step shrink: the equation

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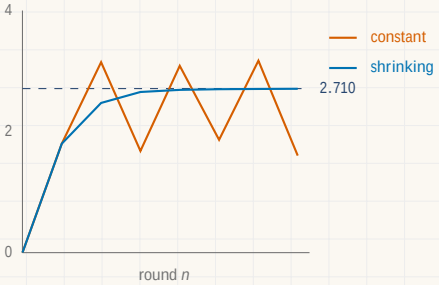

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The answer is $1 + \sqrt[3]{5} = 2.709976$.

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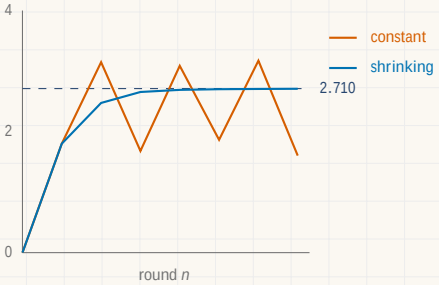

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The answer is $1 + \sqrt[3]{5} = 2.709976$.

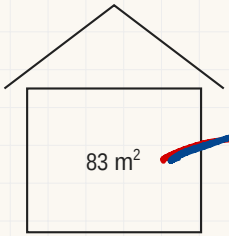
A step that never shrinks has to be lucky. A step that shrinks does not.

3

Section 3

Estimating house prices

One house, one measurement



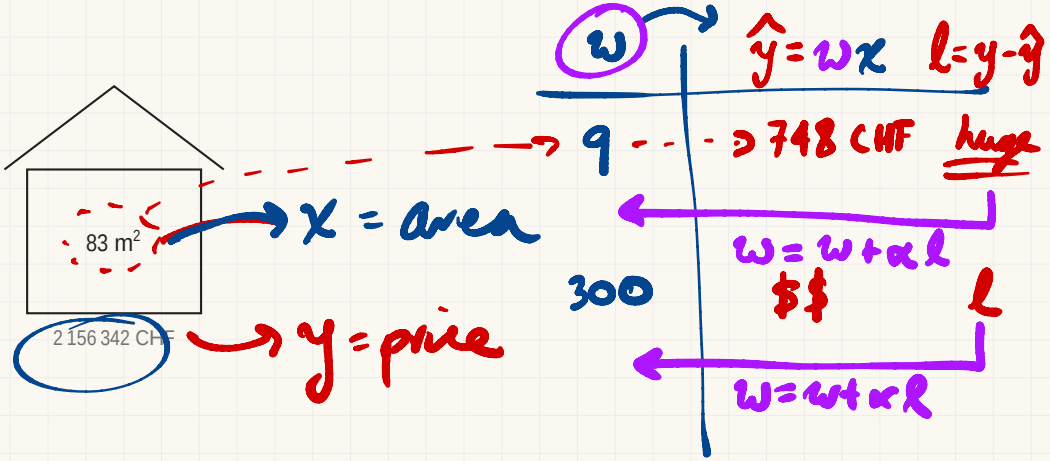
$ is prop to area

x = area

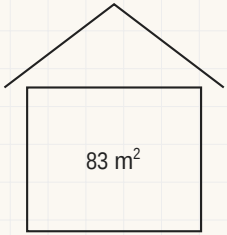
y = price

y = wx

One house, one measurement

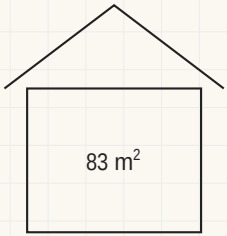


One house, one measurement



Data $x = 83$, the area. $y = 2\,156\,342$, the price.

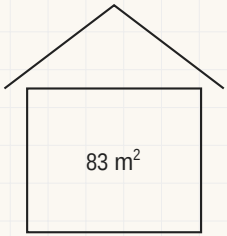
One house, one measurement



Data $x = 83$, the area. $y = 2\,156\,342$, the price.

Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w\,x$.

One house, one measurement

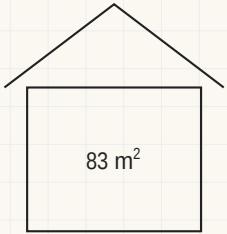


Data $x = 83$, the area. $y = 2\,156\,342$, the price.

Hypothesis Price is proportional to area, so the estimate is $\hat{y} = w\,x$.

Verdict The signed miss $l = y - \hat{y}$.

One house, one measurement



Data	$x = 83$, the area. $y = 2\,156\,342$, the price.
Hypothesis	Price is proportional to area, so the estimate is $\hat{y} = w\,x$.
Verdict	The signed miss $l = y - \hat{y}$.
Problem	Find the w that drives l to zero.

One row and one unknown, so the answer is $y/x = 25\,980.02$ francs per square metre, and the loop that follows never divides.

One house: the loop

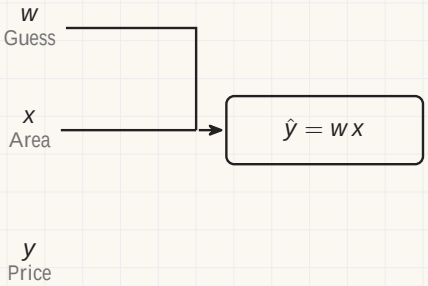
One house: the loop

w
Guess

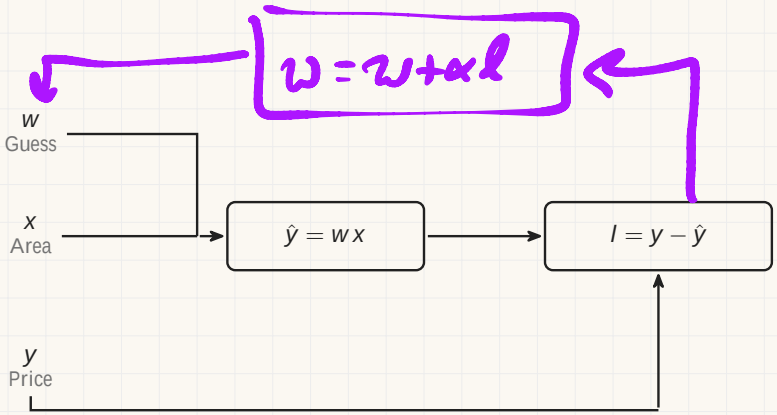
x
Area

y
Price

One house: the loop



One house: the loop



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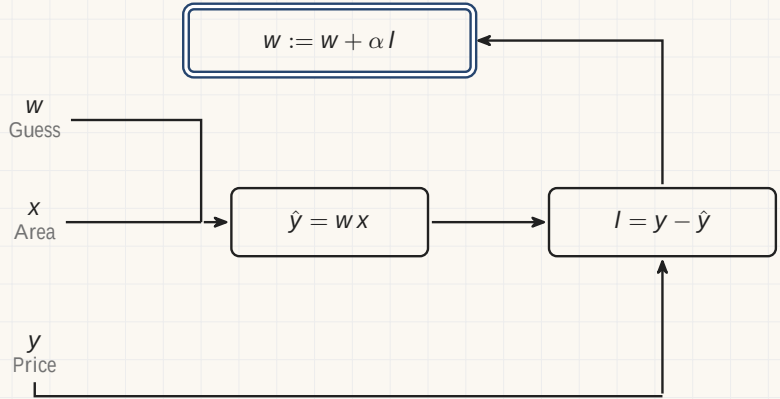
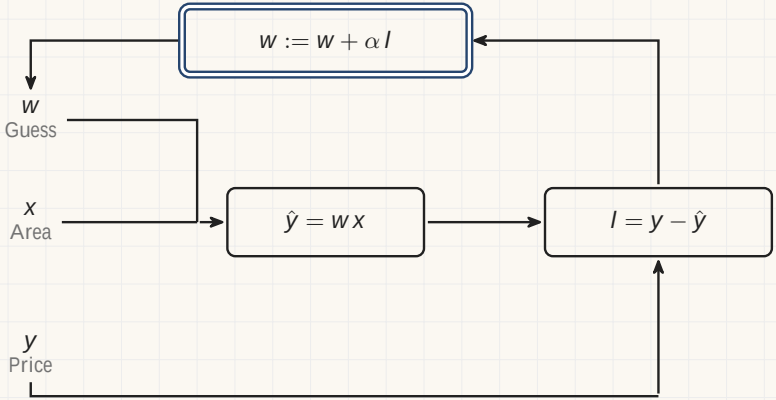


One house: the loop



One house: the algorithm

```
def fit(x, y, w, alpha):
    l = y - w * x              # the verdict: y minus the estimate
    while abs(l) > 0.5:        # more than half a franc out
        w = w + alpha * l      # the update
        l = y - w * x          # score the new guess
    return w
```

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On the house above:

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fit(83, 2156342, 20000, 0.01)
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```

On the house above:

```
fit(83, 2156342, 20000, 0.01)
```

Starting from a guess of 20 000 francs per square metre, this returns 25 980.02 after 8 rounds.

Many houses, one measurement

House	Area	Price

Many houses, one measurement

House	Area	Price
1	83	2 156 342
2	62	1 461 635
3	104	2 650 166
4	128	3 184 899
5	145	3 651 113

House 1 is the house on the last slide. Houses 2 to 5 are synthetic, generated with a □xed seed.

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House	Area	Price
1	83	2 156 342
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Hypothesis The same one, row by row: $\hat{y}_i = w\,x_i$.

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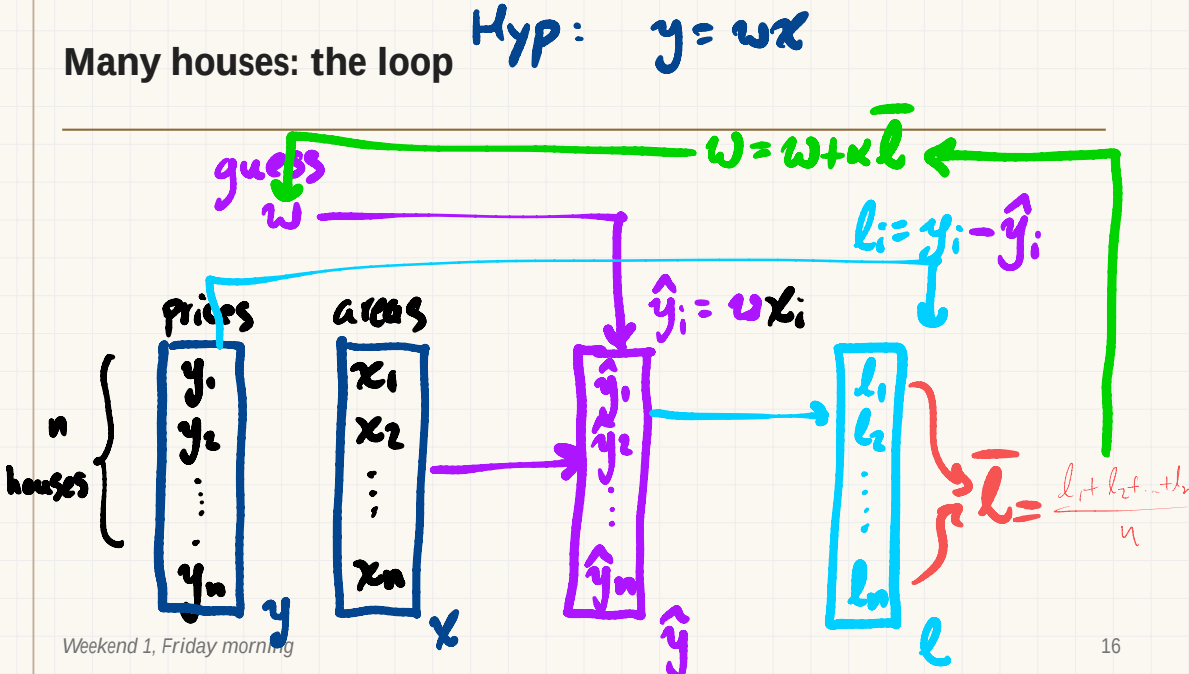
Hypothesis The same one, row by row: $\hat{y}_i = w\,x_i$.

Verdict One miss per row, $l_i = y_i - \hat{y}_i$.

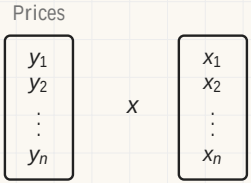
Problem No single w empties every row. Drive the *average* verdict $\bar{l}$ to zero instead.

$$\bar{l} = \frac{l_1 + l_2 + \cdots + l_n}{n}$$

Many houses: the loop



Many houses: the loop



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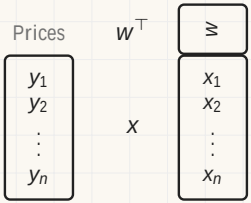

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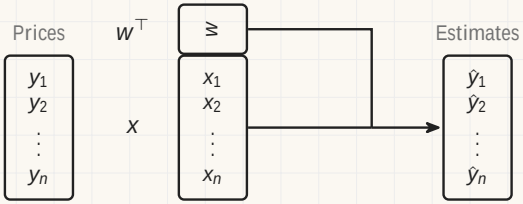

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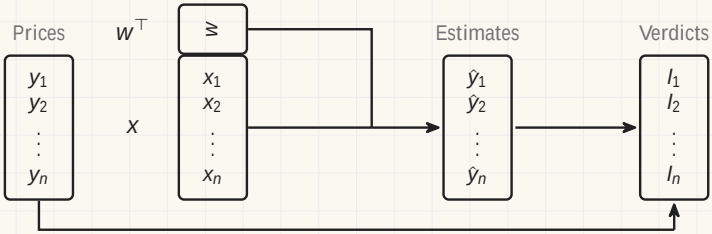

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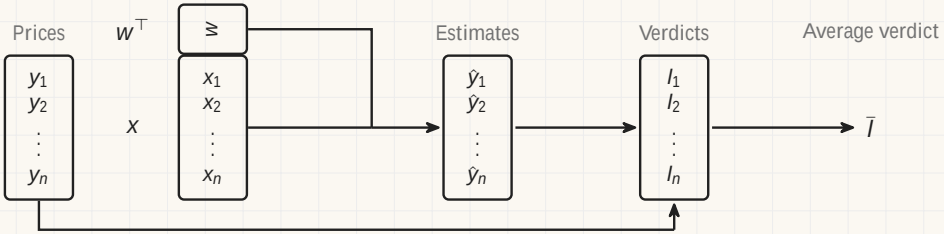
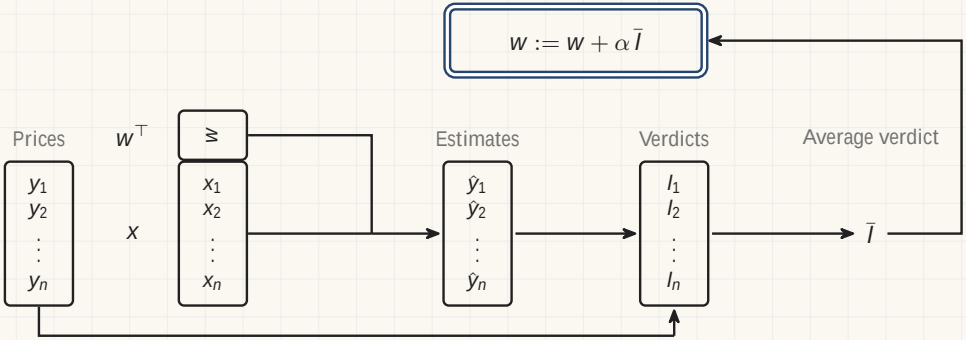


x holds the areas, and $\hat{y}_i = w\,x_i, \quad l_i = y_i - \hat{y}_i, \quad \bar{l} = (l_1 + \cdots + l_n)/n$

Many houses: the loop



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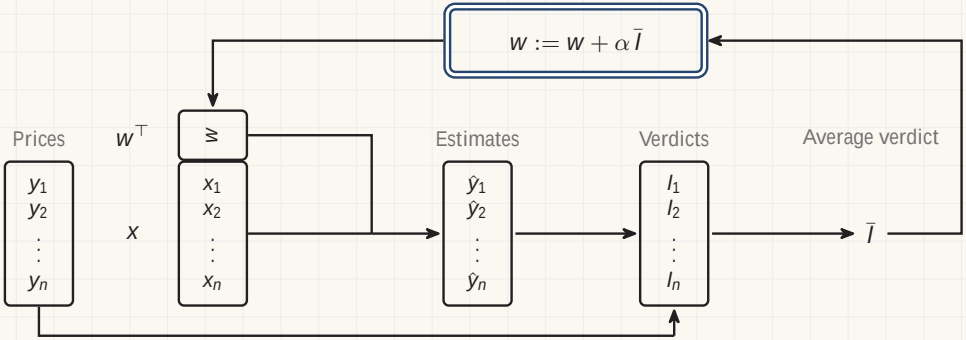

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Many houses: the algorithm

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def verdict(x, y, w):          # the average miss over the rows
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - w * x[i])
    return total / len(x)
def fit(x, y, w, alpha):
    l_bar = verdict(x, y, w)
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    return w
```

The estimate, the verdict and the update are unchanged. Only the averaging is new.

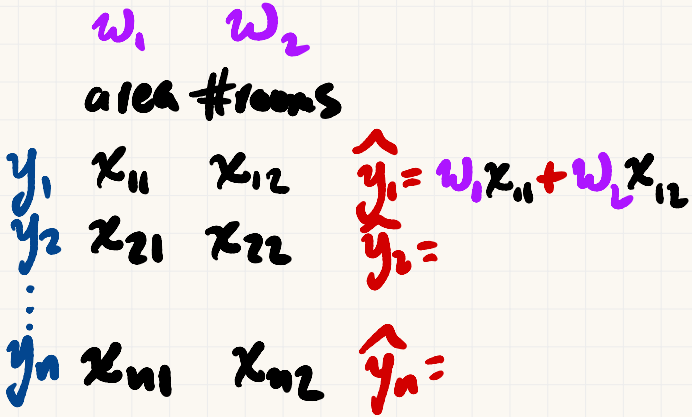
On the five houses, starting from 20 000 with $\alpha = 0.005$, this returns 25 103.74 after 19 rounds.

Two measurements for each house

House	Area	Rooms	Price

Two measurements for each house

House	Area	Rooms	Price
1	83	3.5	2 156 342
2	62	2.5	1 461 635
3	104	4.5	2 650 166
4	128	5.5	3 184 899
5	145	6.5	3 651 113



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Hypothesis Each measurement gets its own weight: $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$.

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Hypothesis Each measurement gets its own weight: $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$.

Verdict Still one number, $\bar{l}$, the average miss.

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House	Area	Rooms	Price
1	83	3.5	2 156 342
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Hypothesis Each measurement gets its own weight: $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$.

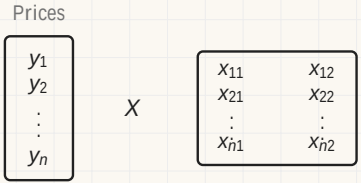
Verdict Still one number, $\bar{l}$, the average miss.

Problem Two unknowns, one verdict. It does not say which weight is at fault.

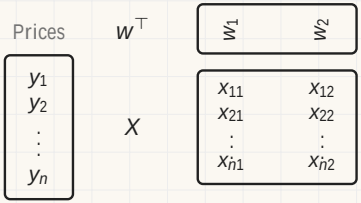
So move one weight at a time.

Two measurements: the loop

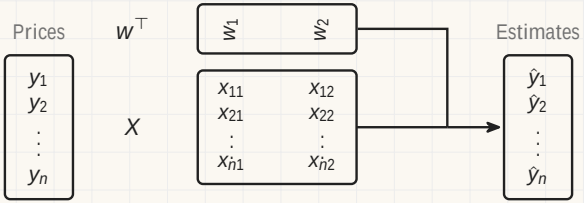
Two measurements: the loop



Two measurements: the loop

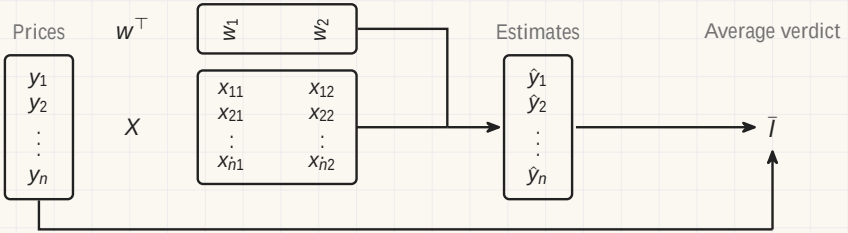


Two measurements: the loop



$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$, and the columns of X are area and rooms.

Two measurements: the loop



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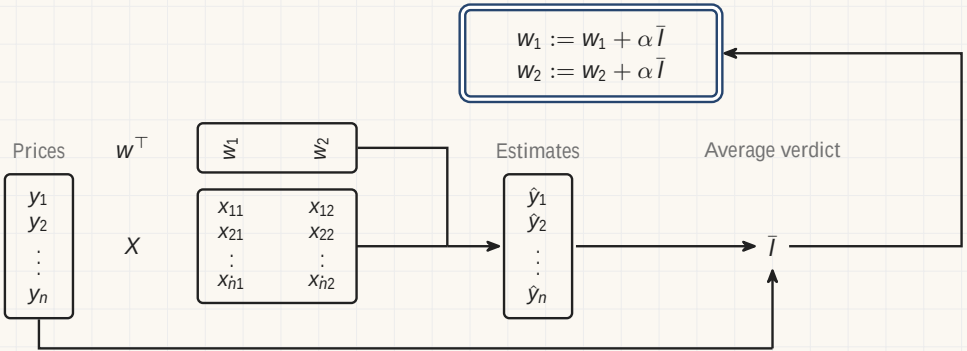
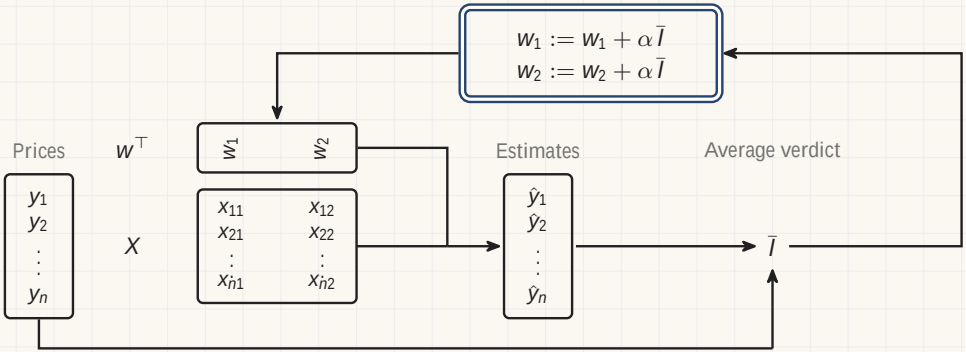


$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2}$, and the columns of X are area and rooms.

One weight moves per round, and they take turns.

Two measurements: the loop



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Two measurements: the algorithm

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def verdict(x, y, w):
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    for i in range(len(x)):
        total = total + (y[i] - w[0] * x[i][0] - w[1] * x[i][1])
    return total / len(x)

def fit(x, y, w, alpha):
    j = 0
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        w[j] = w[j] + alpha * l_bar
        j = 1 - j                    # take turns
        l_bar = verdict(x, y, w)
    return w
```

Two measurements: the algorithm

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        j = 1 - j                    # take turns
        l_bar = verdict(x, y, w)
    return w
```

Starting from $w = (20\,000,\ 0)$ with $\alpha = 0.005$, this returns $(25\,000.71,\ 2\,390.34)$ after 37 rounds.

Many measurements for each house

House	Area	Rooms	Age	Walk	Price

Many measurements for each house

House	Area	Rooms	Age	Walk	Price
1	83	3.5	12	4	2 156 342
2	62	2.5	41	11	1 461 635
3	104	4.5	28	8	2 650 166
4	128	5.5	6	15	3 184 899
5	145	6.5	55	6	3 651 113

Many measurements for each house

House	Area	Rooms	Age	Walk	Price
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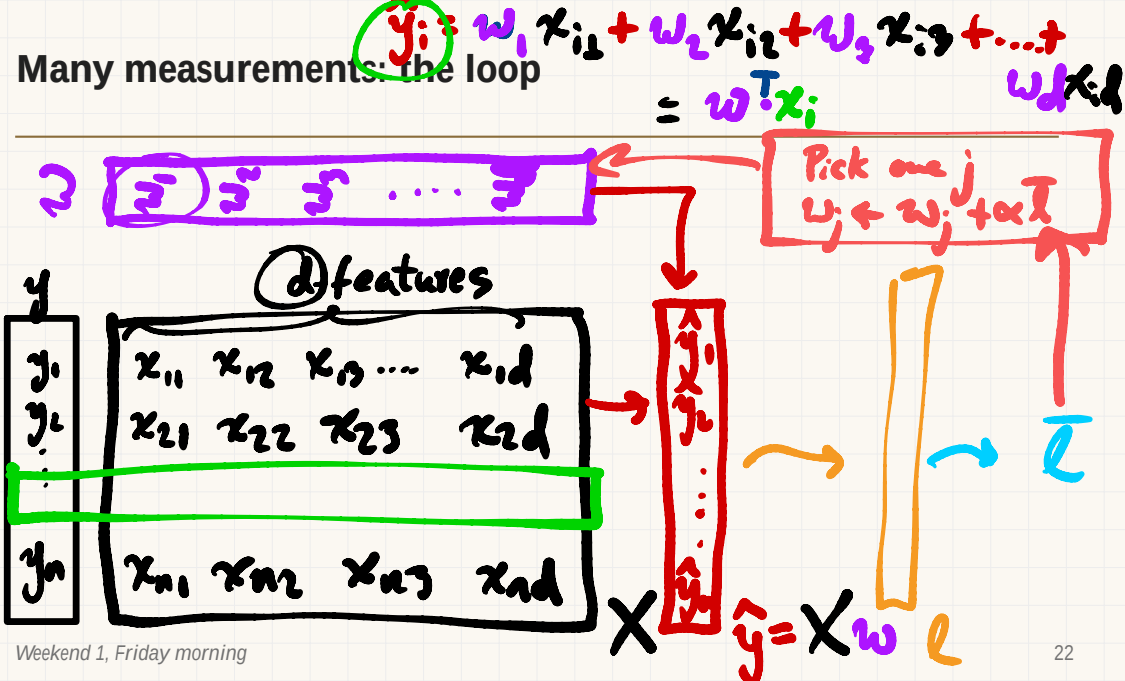
- One verdict, d unknowns. Pick one weight per round and move it.

Many measurements: the loop



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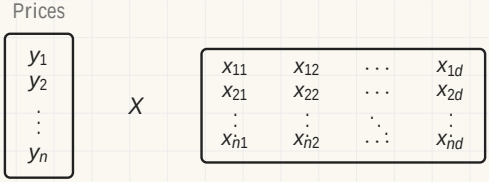

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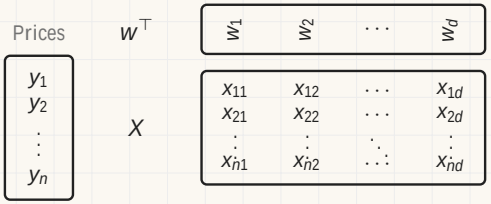

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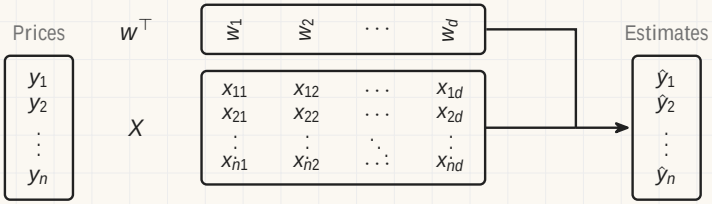

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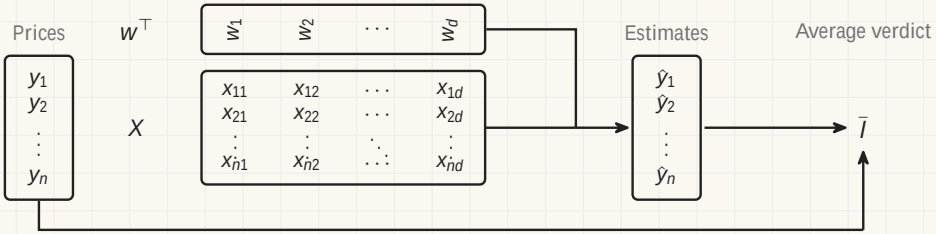

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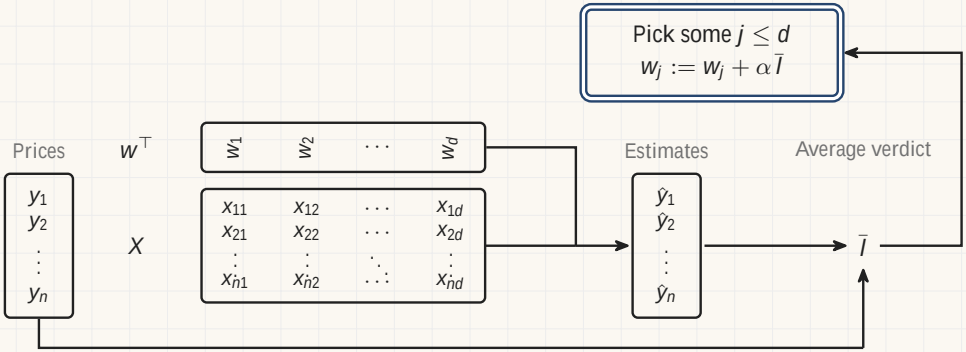

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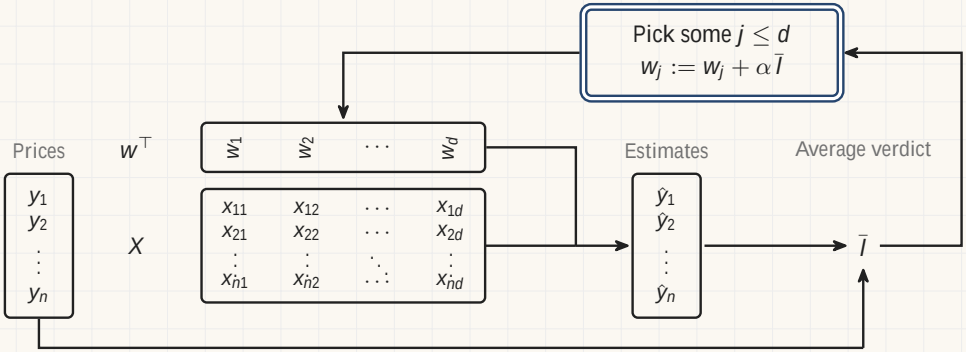

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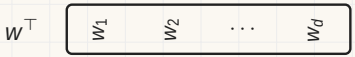

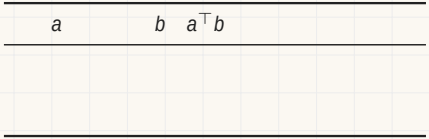

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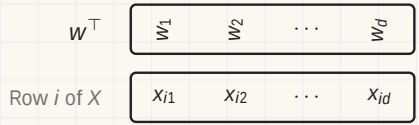
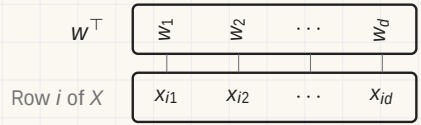


a	b	$a^\top b$

The inner product



$$w^\top x_i = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id}$$

Multiply down each column, then add.

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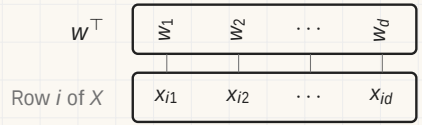

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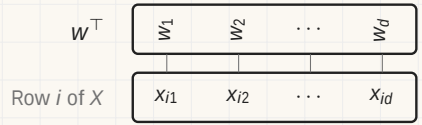

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It also measures similarity.

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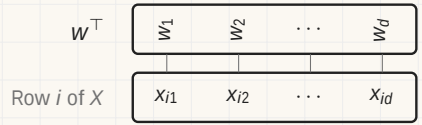

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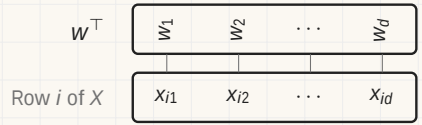

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It scores a document against a query, and one word against another in a language model. It returns all module.

Many measurements: the algorithm

```
def verdict(x, y, w):
    total = 0
    for i in range(len(x)):
        total = total + (y[i] - inner(w, x[i]))
    return total / len(x)
def fit(x, y, w, alpha):
    l_bar = verdict(x, y, w)
    while abs(l_bar) > 0.5:
        j = random.randrange(len(w))   # pick some j
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The three algorithms before it are this one with $d = 1$ and $n = 1$, then $d = 1$, then $d = 2$.

4

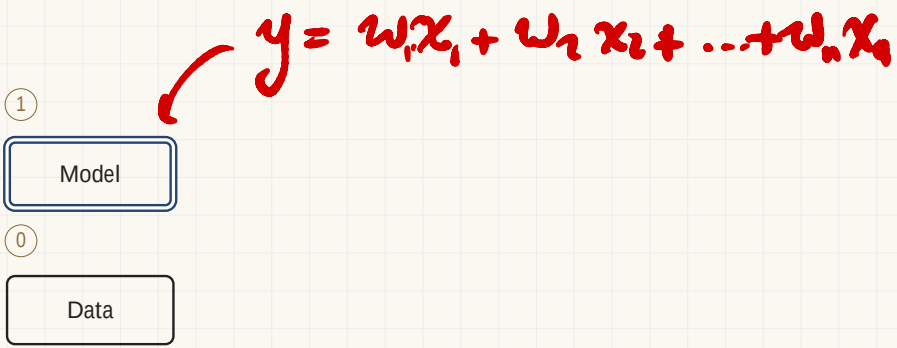
Section 4

In AI

The training loop

The training loop

The training loop



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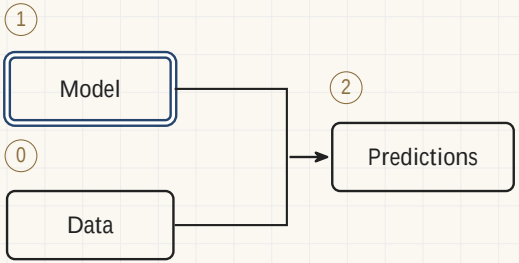

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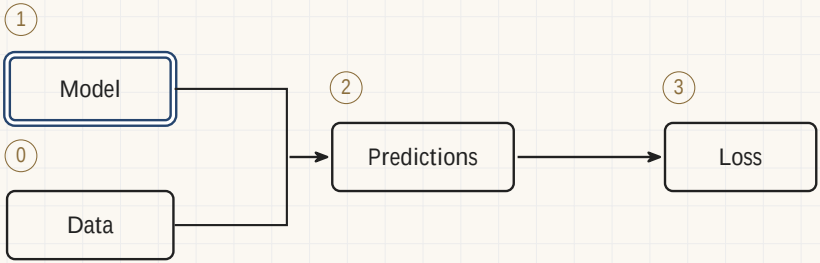

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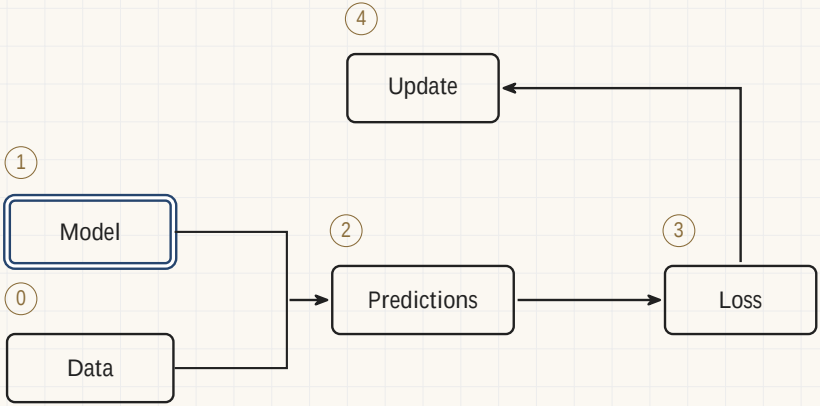

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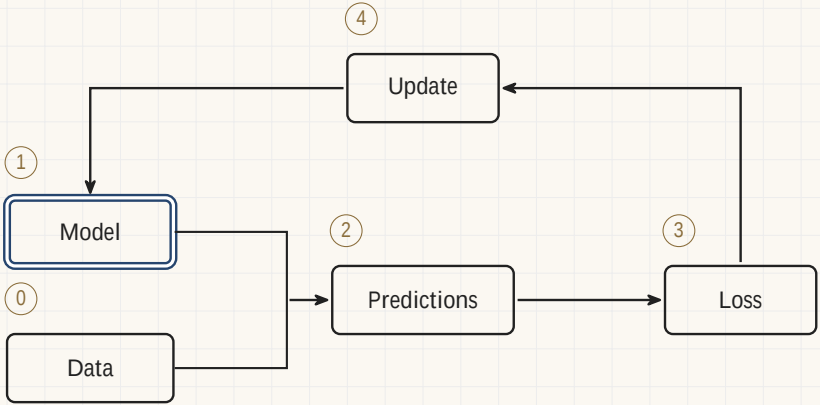

Four boxes and a return arrow, run a few million times.

The morning’s four words, and their AI names

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Guess → Model

The weights *w*

The morning’s four words, and their AI names

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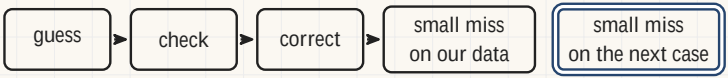
Update	→	**Training step** $w := w + \alpha l$

Statistical learning theory

Vladimir Vapnik



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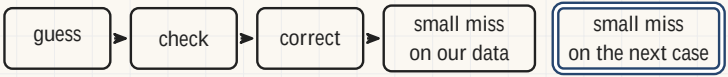

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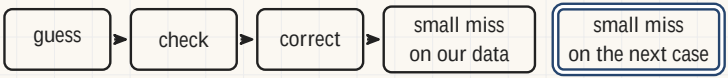

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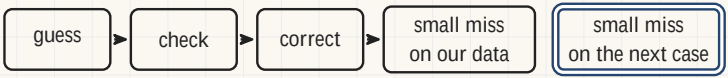

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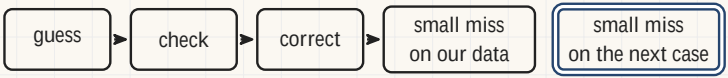

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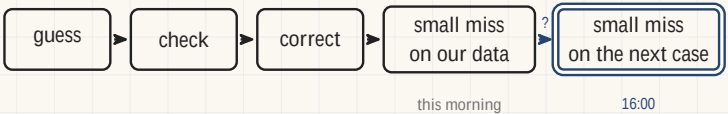

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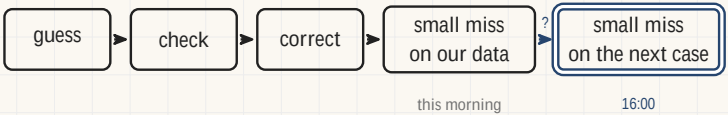

Vapnik prices that arrow: the miss on the next case is at most the miss on our data, plus a penalty that grows with how much the family can bend and shrinks as rows are added. A price rather than a promise.

5

Section 5

Where this ends up

Modern applications

Most modern machine learning and AI applications are □rst-order approximation algorithms.

Family	What its verdict compares

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Four di□erent verdicts. One update rule, and it is the one on the square-root table.